Band-1 Computational Experiment: Beal Conjecture Finite Search

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note is a **finite enumeration probe** (collapse list). It searches $A^x + B^y = C^z$ inside small explicit bounds for triples with $\gcd(A, B, C) = 1$. It does **not** prove the Beal conjecture, does **not** claim the AMS Beal Prize, and does **not** claim Clay Millennium status or Theory of Everything progress. **Problem status: OPEN.**

Abstract

We study Beal's equation

$$A^x + B^y = C^z$$

for positive integers A, B, C and exponents $x, y, z \geq 3$. The Beal conjecture asserts that any such solution forces A, B, C to share a common prime factor (equivalently $\gcd(A, B, C) > 1$). On this box we:

1. Exhaustively enumerate coprime-base pairs with bases $A, B, C \leq 50$ and exponents in $\{3, \ldots, 10\}$, looking up sums in a precomputed power table.
2. Stretch the same exact method to bases ≤ 200 (exponents 3..12) and an optimized Norvig-style scan bases ≤ 500 (exponents odd primes $+4$, ≤ 30).
3. Ledger literature status: conjecture **OPEN**; AMS $1,000,000 Beal Prize unclaimed; prior computational searches report no counterexamples to much larger bounds (not re-proved here).

Box result: 0 counterexamples in all three scans. Conjecture remains **OPEN**.

Problem statement

Beal's conjecture (Beal, 1993; announced *Notices AMS* Dec 1997). If $A^x + B^y = C^z$ with A, B, C, x, y, z positive integers and $x, y, z > 2$, then A, B, and C have a common prime factor.

A **counterexample** is a solution with $\gcd(A, B, C) = 1$. Fermat's Last Theorem (Wiles) is the special case $x = y = z = n > 2$.

This experiment. Exact finite search inside box-feasible bounds, plus a literature ledger. No infinitary claim and no prize claim are asserted.

Literature ledger (not re-proved on box)

Item	Status	Note
Beal conjecture	**OPEN**	No refereed proof or verified counterexample

Item	Status	Note
AMS Beal Prize	**$1,000,000 unclaimed**	Held by AMS; requires refereed publication + (for cex) independent verification
Norvig computational search	Literature	No cex for $A, B \leq 1000$ with exponents up to 100 (optimized)
Jarnicki–Konerding (cited by Norvig)	Literature	A, B up to 200,000, exponents up to 5,000 — no cex
Consequence	Heuristic	Any counterexample, if one exists, must be **large**

One-liner. Beal conjecture remains **OPEN**; AMS Beal Prize $1M unclaimed; known searches found no counterexamples to large bounds, so any counterexample must be large.

Critical distinction. A search to bases 50 (or even literature 10^5) is a **finite certificate**, not a proof of the conjecture.

Method (box)

1. Precompute all powers A^e for $A \in \{1, \ldots, M\}$ and exponents in range (primary: all $e \in \{3, \ldots, 10\}$).
2. Build a reverse map $\{C^z \mapsto (C, z)\}$.
3. For every unordered pair of bases with $\gcd(A, B) = 1$, test every exponent pair; if $A^x + B^y$ hits the reverse map at (C, z), check $\gcd(A, B, C)$.
4. Filtering $\gcd(A, B) = 1$ is sound for counterexample search: if $\gcd(A, B) = p > 1$ then $p \mid A^x + B^y = C^z$, hence $p \mid C$, so $\gcd(A, B, C) \geq p > 1$ (not a counterexample). Same-base sums likewise force a common factor.
5. Sanity scan (no $\gcd(A, B)$ filter) recovers known common-factor identities such as $3^3 + 6^3 = 3^5$.

Arithmetic. Python arbitrary-precision `int` only (no floats).

Results (this box, 2026-09-11)

Scan	Bases	Exponents	Counterexamples	Wall (part of ~1 s total)
Primary	$\leq$ **50**	**all** 3..10	**0**	included
Secondary	≤ 200	all 3..12	**0**	included
Stretch (Norvig-style)	≤ 500	odd primes +4, ≤ 30	**0**	included
Unit tests	—	—	**PASS**	—
Sanity common-factor hits (primary window)	≤ 50, 3..10	—	**17** non-cex identities (e.g. $3^3 + 6^3 = 3^5$)	—

Primary coprime power-combinations checked: **46{,}728**. Total wall time ≈ 1.0 s (well inside the 2–3 min budget).

Classic non-counterexample (sanity)

$$3^3 + 6^3 = 27 + 216 = 243 = 3^5, \qquad \gcd(3, 6, 3) = 3 > 1.$$

This satisfies the equation but **not** the coprimality condition required of a counterexample — consistent with the conjecture.

Honesty verdict

Finite enumeration only. Exact search in the stated bounds found **zero** counterexamples and recovered known common-factor solutions as a sanity check. Literature computational bounds extend far beyond this box but likewise do not close the problem. **Beal's conjecture remains OPEN.** This note does **not** claim Beal solved, does **not** claim the AMS Beal Prize, and does **not** claim Clay / ToE.

Reproducibility

```
cd /workspace/experiments/band1/beal
python3 beal_search.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `README.md`.

References (selected)

1. D. Andrew Beal; announcement in *Notices Amer. Math. Soc.* **44** (1997), Dec issue (Beal Prize / conjecture).
2. American Mathematical Society, *Beal Prize Rules and Procedures*, https://www.ams.org/profession/prizes-awards/ams-supported/beal-prize-rules (accessed 2026-09-11).
3. R.D. Mauldin, *The Beal Conjecture and Prize*, https://math.unt.edu/~mauldin/beal.html.
4. P. Norvig, *Beal's Conjecture Revisited*, https://www.norvig.com/beal.html (2015–2017); cites Jarnicki–Konerding large-bound search.
5. A. Wiles, Modular elliptic curves and Fermat's Last Theorem, *Ann. of Math.* **141** (1995), 443–551.